

Appendix to the article

“Improving accuracy of age estimates for insect evidence – calibration of physiological age at emergence (k) using insect size but without “ k versus size” model”

The negative correlation between the “true k ” and size

A negative correlation of ‘true k ’ and size seems to be a strong rule (see discussion section of the manuscript). However, if there are good reasons to claim that particular species is characterized by a positive correlation (e.g. a parasitoid species is analyzed), our formula could be simply adjusted to the following form (the first subtraction changes to addition, “-“changes into “+“sign):

$$K_c^+(l) = k + \frac{s_e \sqrt{3m}}{l_{max} - l_{min}} \left(l - \frac{l_{max} + l_{min}}{2} \right).$$

The standard deviation of the “true k ”

Most statistical software packages (like R language or Statistica) calculate standard error of intercept (regression coefficient) by the formula:

$$s_e = \frac{sd(e)}{\sqrt{m}} \cdot \sqrt{1 + \frac{\bar{x}^2}{Var(x)}}$$

where s_e denotes standard error of intercept, $sd(e)$ denotes standard deviation of errors ($e_i = y_i - \hat{y}_i$, i.e. the differences between observed data and predictions), m denotes number of observations, $\bar{x}$ and $Var(x)$ denote mean and variance of independent variable, respectively.

In case of development model and Ikemoto and Takai’s method we should remember that:

$$sd(e) = sd(DT - t_0 D) \approx sd(k_t)$$

The equality is true because variance is translation invariant, and estimation stems from the formula of the law of total effective temperatures.

Now formula of standard error of k is as follows:

$$s_e \approx \frac{sd(k_t)}{\sqrt{m}} \cdot \sqrt{1 + \frac{\bar{D}^2}{Var(D)}}$$

where s_e denotes standard error of estimate of k , D denotes time of insect development and m denotes number of temperature data points used to derive a development model. Elementary algebra gives:

$$sd(k_t) \approx s_e \sqrt{m} \cdot \left(\sqrt{1 + \frac{\bar{D}^2}{Var(D)}} \right)^{-1}$$

Factors s_e and m are known, but term $\left(\sqrt{1 + \frac{\bar{D}^2}{Var(D)}} \right)^{-1}$ is impossible to calculate without a dataset.

However, in case of Ikemoto and Takai’s method time of development is treated like a deterministic value, and it is strictly associated with levels of temperature, which were studied by the experimenter. Due to these reasons and lack of a better solution we can estimate mean and variance as if the time of development was uniformly distributed. Then:

$$\sqrt{1 + 3 \frac{(D_{min} + D_{max})^2}{(D_{min} - D_{max})^2}}$$

Now it is easy to notice that:

$$\begin{aligned} \sqrt{1 + 3 \frac{(D_{min} + D_{max})^2}{(D_{min} - D_{max})^2}} &< 2 \\ \frac{(D_{min} + D_{max})^2}{(D_{min} - D_{max})^2} &< 1 \\ \frac{D_{min} + D_{max}}{D_{max} - D_{min}} &< 1 \\ D_{min} &< 0 \end{aligned}$$

which means that a maximal natural value of the considered term is 2. So, the formula:

$$sd(\mathbf{k}_t) \approx \frac{s_e \sqrt{m}}{2}$$

provides a maximal reasonable correction for a constant k .

In practice, if in a specific case examiners possess better information about mean and variance of the time of development, for instance a dataset that was used to derive the thermal summation model; we recommend using them and the full formula for $sd(\mathbf{k}_t)$ to obtain better results.

The standard deviation of size (i.e. L)

Because of practical limitations, the length of an insect is usually measured with half-millimeter accuracy. Assuming continuity and according to the transformation done in the “Results” section (which implies that every value contains one observation), values of this variable form an arithmetic sequence. Therefore, the sample of insect lengths is assumed to be an arithmetic sequence. For this reason, knowledge about maximal and minimal length values is enough to calculate the standard deviation of length.

Let us take an arithmetic sequence $\mathbf{a}$ with m elements, i.e.,

$$\begin{aligned} \mathbf{a} &= (a_1, a_2, \dots, a_{m-1}, a_m) \\ a &= a_1 < a_2 < a_3 < \dots < a_{m-1} < a_m = b \\ \forall i \in ([1, m] \cap \mathbb{N}): a_{i+1} - a_i &= \frac{b - a}{m - 1} = \text{const.} \end{aligned}$$

The standard deviation of a sequence $\mathbf{a}$ is given by the following formula:

$$s(\mathbf{a}) = \sqrt{\frac{1}{m} \sum_{i=1}^m (a_i - \bar{a})^2}.$$

Notice that $s^2(\mathbf{x})$ is invariant with respect to the changes in location parameter and is scaled by the square of constant when all values are scaled by this constant, i.e.,

$$s^2(\alpha \mathbf{a} + \beta) = \alpha^2 \cdot s^2(\mathbf{a}).$$

Since $\mathbf{a}$ is an arithmetic sequence, we have

$$\begin{aligned} \mathbf{a} &= \left(a, a + \frac{b - a}{m - 1}, a + 2 \frac{b - a}{m - 1}, \dots, b \right) = \\ &= \left(a, a + \frac{b - a}{m - 1}, a + 2 \frac{b - a}{m - 1}, \dots, a + (m - 1) \frac{b - a}{m - 1} \right) = \end{aligned}$$

$$= (a, \dots, a) + \frac{b-a}{m-1} (0, 1, \dots, m-1)$$

Let $\ddot{\mathbf{a}} = (0, 1, \dots, m-1)$. Then

$$s^2(\mathbf{a}) = \left(\frac{b-a}{m-1} \right)^2 \cdot s^2(\ddot{\mathbf{a}}),$$

we have $x_1 = 0, x_2 = 1, \dots, x_n = m-1$, and

$$\bar{x} = \frac{1}{m} \sum_{i=1}^{m-1} i = \frac{1}{m} \cdot \frac{1+m-1}{2} \cdot (m-1) = \frac{m-1}{2}$$

since

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$$

and

$$s^2(\mathbf{x}) = \frac{1}{n} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right),$$

we have

$$\begin{aligned} s^2(\ddot{\mathbf{a}}) &= \frac{1}{m} \left(\sum_{i=1}^n i^2 - m \left(\frac{m-1}{2} \right)^2 \right) = \\ &= \frac{1}{m} \left(\frac{1}{6} m(m+1)(2m+1) - m \left(\frac{m-1}{2} \right)^2 \right) = \\ &= \left(\frac{1}{6} (2m-1) - \frac{m-1}{4} \right) (m-1) = \\ &= \frac{1}{12} (m^2 - 1). \end{aligned}$$

Finally

$$s^2(\mathbf{a}) = \frac{1}{12} (m^2 - 1) \left(\frac{b-a}{m-1} \right)^2 = \frac{(m+1)(b-a)^2}{12(m-1)}.$$

Therefore, the standard deviation of the length, which can be easily used in practice, is as follows:

$$sd(\mathbf{l}) = \frac{(l_{\max} - l_{\min})}{2} \sqrt{\frac{m+1}{3m-3}}. \quad (1)$$

However, to simplify K_c formula, we can use trivial fact from the theory of limits that

$$\lim_{n \rightarrow \infty} \sqrt{\frac{m+1}{3m-3}} = \frac{\sqrt{3}}{3}.$$

Then for a sufficiently large m , we have

$$sd(\mathbf{l}) \approx \frac{\sqrt{3}(l_{\max} - l_{\min})}{6}. \quad (2)$$

It is much easier to use (2) than (1) and the difference between them is negligible in practical applications.